Algebraic Notation and Substitution

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KEY WORDS variable *(taurangi)* · expression *(kīanga taurangi)* · substitute *(whakakapi)* · formula *(tātai)*

A Algebraic notation ●○○

Remember: In algebra the × sign is left out and the number goes first: $a \times 3 = 3a$. Division is written as a fraction: $a \div 4 = \frac{a}{4}$.

1. Write each expression the short way.

a) $3 \times a$ = _____ **b)** $b \times 5$ = _____ **c)** $a \times b$ = _____ **d)** $a \times a$ = _____

e) $x \div 4$ = _____ **f)** $2 \times a \times b$ = _____ **g)** $y \times y \times y$ = _____ **h)** $1 \times n$ = _____

i) $7 \times m$ = _____ **j)** $p \div 2$ = _____ **k)** $n \times 6$ = _____ **l)** $3 \times a + 2 \times b$ = _____

B Writing expressions ●○○

Watch out: "6 less than n" is $n - 6$, not $6 - n$. The order matters when you subtract or divide.

2. Write an expression for each statement.

a) 5 more than n _____ **b)** n multiplied by 4 _____

c) 6 less than n _____ **d)** n divided by 3 _____

e) double n _____ **f)** n squared _____

g) double n, then add 1 _____ **h)** subtract n from 10 _____

C Expressions for situations ●●○

Remember: Choose the operation that fits the story, then write it the short way. "3 dollars each for n pens" is $3 \times n$, written $3n$.

3. Write an expression for each situation.

a) The cost, in dollars, of n pens at \$3 each

b) The number of lollies left when you start with 20 and eat n of them

c) The age of Mia, who is 4 years older than her brother. Her brother is n years old.

d) The share each person gets when n lollies are shared equally between 5 people

e) The cost, in dollars, of a taxi that charges \$4 plus \$2 for each of n kilometres

f) The perimeter of a square with sides of n cm

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D Substituting into an expression ●●○

Remember: Replace each letter with its number, then calculate. If $a = 3$, then $2a = 2 \times 3 = 6$ (not 23).

4. Find the value of each expression when $a = 3$ and $b = 5$.

a) $a + b$ = _____ **b)** $2a$ = _____ **c)** ab = _____ **d)** $3b - a$ = _____

e) a^2 = _____ **f)** $2a + 3b$ = _____ **g)** $10 - 2a$ = _____ **h)** $4b$ = _____

5. Find the value of each expression when $x = 6$ and $y = 2$.

a) $x - y$ = _____ **b)** $5y$ = _____ **c)** $\frac{x}{y}$ = _____ **d)** xy = _____

e) $x + 3y$ = _____ **f)** y^2 = _____ **g)** $2x - y$ = _____ **h)** $\frac{x}{2}$ = _____

E Substituting into a formula ●●●

6. Use each formula.

a) The area of a rectangle is $A = lw$. Find A when $l = 8$ and $w = 5$.

b) The perimeter of a square is $P = 4l$. Find P when $l = 7$.

c) The cost in dollars of hiring a hall for n hours is $C = 5n + 20$. Find C when $n = 6$.

d) The perimeter of a rectangle is $P = 2l + 2w$. Find P when $l = 7$ and $w = 4$.

e) The distance a car travels in t hours is $d = 60t$. Find d when $t = 3$.

f) The number of matchsticks in pattern n is $T = 3n + 2$. Find T when $n = 10$.

★ Extension

7. Try different values of n to find the one that makes $3n + 1 = 16$ true.

8. Find the value of each expression when $a = 2$ and $b = 5$.

a) $a^2 + b^2$ **b)** $3(a + b)$ **c)** $\frac{4b}{a - 1}$

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Algebraic Notation and Substitution

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS variable *(taurangi)* · expression *(kīanga taurangi)* · substitute *(whakakapi)* · formula *(tātai)*

A Algebraic notation ●○○

Remember: In algebra the × sign is left out and the number goes first: $a \times 3 = 3a$. Division is written as a fraction: $a \div 4 = \frac{a}{4}$.

1. Write each expression the short way.

a) $5 \times a$ = _____ **b)** $c \times 7$ = _____ **c)** $x \times y$ = _____ **d)** $b \times b$ = _____

e) $y \div 3$ = _____ **f)** $4 \times p \times q$ = _____ **g)** $a \times a \times a$ = _____ **h)** $1 \times m$ = _____

i) $9 \times k$ = _____ **j)** $t \div 5$ = _____ **k)** $n \times 8$ = _____ **l)** $2 \times x + 5 \times y$ = _____

B Writing expressions ●○○

Watch out: "6 less than n" is $n - 6$, not $6 - n$. The order matters when you subtract or divide.

2. Write an expression for each statement.

a) 8 more than n _____ **b)** n multiplied by 6 _____

c) 4 less than n _____ **d)** n divided by 5 _____

e) triple n _____ **f)** n cubed _____

g) triple n, then subtract 2 _____ **h)** subtract n from 20 _____

C Expressions for situations ●●○

Remember: Choose the operation that fits the story, then write it the short way. "3 dollars each for n pens" is $3 \times n$, written $3n$.

3. Write an expression for each situation.

a) The cost, in dollars, of n books at \$8 each

b) The number of pages left when a book has 50 pages and you have read n of them

c) The age of Tom, who is 6 years older than his sister. His sister is n years old.

d) The share each person gets when n grapes are shared equally between 4 people

e) The cost, in dollars, of a taxi that charges \$5 plus \$3 for each of n kilometres

f) The perimeter of an equilateral triangle with sides of n cm

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D Substituting into an expression ●●○

Remember: Replace each letter with its number, then calculate. If $a = 3$, then $2a = 2 \times 3 = 6$ (not 23).

4. Find the value of each expression when $a = 4$ and $b = 6$.

a) $a + b$ = _____ **b)** $2a$ = _____ **c)** ab = _____ **d)** $3b - a$ = _____

e) a^2 = _____ **f)** $2a + 3b$ = _____ **g)** $10 - 2a$ = _____ **h)** $4b$ = _____

5. Find the value of each expression when $x = 8$ and $y = 2$.

a) $x - y$ = _____ **b)** $5y$ = _____ **c)** $\frac{x}{y}$ = _____ **d)** xy = _____

e) $x + 3y$ = _____ **f)** y^2 = _____ **g)** $2x - y$ = _____ **h)** $\frac{x}{2}$ = _____

E Substituting into a formula ●●●

6. Use each formula.

a) The area of a rectangle is $A = lw$. Find A when $l = 9$ and $w = 6$.

b) The perimeter of a square is $P = 4l$. Find P when $l = 9$.

c) The cost in dollars of hiring a boat for n hours is $C = 8n + 15$. Find C when $n = 5$.

d) The perimeter of a rectangle is $P = 2l + 2w$. Find P when $l = 8$ and $w = 3$.

e) The distance a car travels in t hours is $d = 80t$. Find d when $t = 4$.

f) The number of matchsticks in pattern n is $T = 4n + 1$. Find T when $n = 10$.

★ Extension

7. Try different values of n to find the one that makes $4n - 3 = 17$ true.

8. Find the value of each expression when $a = 3$ and $b = 4$.

a) $a^2 + b^2$ **b)** $3(a + b)$ **c)** $\frac{4b}{a - 1}$

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